

ClickPost Intent Capture AI Platform

Project Memo

Author Name: Prasanna Kumar

Project Title: ClickPost Intent Capture AI Platform

Technology Stack: Python, FastAPI, Streamlit, Pandas, Docker, Render, Groq LLM, Matplotlib

Project Type: AI-Powered Buying Intent Detection Platform

1. Project Overview

The ClickPost Intent Capture AI Platform is an AI-powered application that helps sales and business development teams identify companies that are likely to purchase logistics or shipping services. Instead of manually searching through multiple news sources and online platforms, the system automatically collects buying intent signals, analyzes them, ranks companies based on their likelihood of becoming customers, and generates personalized outreach messages using a Large Language Model (LLM).

The platform consists of two major components: a FastAPI Backend for processing data and exposing REST APIs, and a Streamlit Dashboard for interactive visualization and business insights. The entire application is containerized using Docker and deployed to cloud platforms, making it production-ready.

2. Problem Statement

Sales teams spend a significant amount of time manually identifying potential customers. Information about company expansion, hiring, funding, product launches, and partnerships is scattered across different online sources, making lead generation slow, inconsistent, and difficult to scale. The objective of this project is to automate this entire workflow using Artificial Intelligence and provide a ranked list of high-intent companies with personalized outreach content.

3. Project Objectives

- Automatically collect buying intent signals
- Process and analyze company information
- Rank companies based on intent score
- Generate AI-powered personalized emails and LinkedIn messages
- Provide an interactive analytics dashboard
- Deploy the application on cloud infrastructure

4. System Architecture

The platform follows a modular pipeline: company data is loaded from the dataset; intent signals are collected from online sources, cleaned, and processed; companies are ranked using a scoring engine; AI generates personalized emails and LinkedIn messages; FastAPI serves the processed data through REST APIs; Streamlit consumes these APIs to display interactive dashboards; Docker packages the application for deployment; and Render hosts the backend API while Streamlit Community Cloud hosts the frontend dashboard.

5. Technology Stack

Language: Python

Backend: FastAPI, Uvicorn

Frontend: Streamlit

Data Processing: Pandas, NumPy

Visualization: Matplotlib

AI & LLM: Groq API

APIs: Requests, Feedparser

Deployment: Docker, Render, Streamlit Community Cloud

Version Control: Git, GitHub

6. Project Workflow

Load company data from CSV → collect buying intent signals → save processed signals → calculate company scores → generate company rankings → produce personalized outreach content → save output files → expose APIs using FastAPI → display insights in Streamlit → allow CSV downloads and filtering. This workflow enables complete automation of lead discovery and outreach generation.

7. Key Features

- Automated intent signal collection and AI-based lead ranking
- Company score visualization and interactive Streamlit dashboard
- Personalized AI-generated emails and LinkedIn outreach generation
- CSV export functionality and REST API integration
- Dockerized deployment and cloud hosting

8. Deployment

Backend: FastAPI, Docker, Render. Frontend: Streamlit Community Cloud. The frontend communicates with the backend through REST APIs, allowing the dashboard to fetch real-time ranking, signals, and outreach data.

9. Challenges Faced

During development, several deployment and integration challenges were encountered, including Docker image configuration, missing CSV files during deployment, Git ignore configuration issues, incorrect API paths, Streamlit deployment errors, Render file path management, and empty API responses. Each issue was resolved through proper project restructuring, Docker optimization, improved API validation, and better deployment practices.

10. Project Outcomes

The final system successfully collects buying intent signals automatically, produces ranked company lists, generates AI-powered personalized outreach, displays interactive dashboards, supports CSV downloads, runs successfully using Docker containers, and is deployed on cloud infrastructure. The project demonstrates practical skills in AI engineering, backend development, frontend visualization, cloud deployment, and production-ready software design.

11. Future Enhancements

Real-time news monitoring, database integration (PostgreSQL), user authentication, CRM integration (Salesforce, HubSpot), scheduled data pipelines, advanced machine learning scoring models, multi-agent AI workflows, analytics and reporting dashboard, email automation, and Kubernetes deployment.

12. Conclusion

The ClickPost Intent Capture AI Platform demonstrates how Artificial Intelligence can automate lead generation and sales intelligence. By integrating data collection, intent analysis, AI-generated outreach, REST APIs, and interactive dashboards into a single workflow, the platform significantly reduces manual effort while improving sales efficiency. This project showcases practical expertise in Python, FastAPI, Streamlit, Docker, cloud deployment, and AI integration, making it a strong portfolio project for Machine Learning Engineer, AI Engineer, and Data Scientist roles.